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Effects of resistance training on glycaemic control in adults with type 2 diabetes mellitus: a narrative review of mechanisms, clinical evidence, and practical application

Running title: Resistance training and glycaemic control in T2DM

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Abstract

Background. Skeletal muscle is the main site of insulin-mediated glucose disposal, and its loss aggravates insulin resistance in type 2 diabetes mellitus (T2DM).

Aim. To synthesise evidence on the mechanisms, efficacy, comparative and combined effects, training parameters, safety and prescription of resistance training for glycaemic control in adults with T2DM.

Material and methods. A narrative review with a structured search of PubMed/MEDLINE, Scopus and the Cochrane Library (2007-2026), prioritising randomised trials, systematic reviews, meta-analyses and guidelines.

Results. Resistance training improves glycaemic control by increasing or preserving muscle mass, muscle glucose uptake and insulin sensitivity. Meta-analyses report average HbA1c reductions of 0.3-0.6 percentage points, comparable to aerobic training, though findings vary; combined training generally provides the greatest benefit. Evidence on dose is inconsistent;

intensity relates to benefit more than volume. Strength gains are consistent; lean-mass effects less so. In one randomised trial, adverse events were mainly musculoskeletal and more frequent than in controls.

Conclusions. Resistance training is effective, generally safe and guideline-endorsed. It should be undertaken 2-3 days per week, individualised, progressed gradually and, where feasible, combined with aerobic exercise. Implementation requires clinician endorsement, referral pathways and attention to adherence in older and higher-risk adults.

Keywords: resistance training; strength training; type 2 diabetes mellitus; glycaemic control; HbA1c; insulin sensitivity

1. Introduction

Type 2 diabetes mellitus (T2DM) is one of the most common chronic diseases worldwide, and its rising prevalence places a growing burden on individuals and healthcare systems, driven by population ageing, urbanisation, obesity, and physical inactivity [1]. T2DM is characterised by insulin resistance together with progressive failure of the compensatory mechanisms of pancreatic beta cells, resulting in chronic hyperglycaemia. Over time, hyperglycaemia causes microvascular complications (retinopathy, nephropathy, and neuropathy) and macrovascular complications (cardiovascular disease, peripheral vascular disease, and lower-limb amputation) [1,2], which account for the substantial morbidity, mortality, and healthcare costs associated with the condition. Global epidemiological data confirm that the burden of T2DM has risen steadily since 1990 and - according to projections accounting for population growth, population ageing, and lifestyle changes - will continue to increase until at least 2044 [1].

Glycated haemoglobin (HbA1c), which reflects average glycaemia over the preceding two to three months, is the principal measure of long-term glycaemic control and one of the key determinants of complication risk. Pooled evidence from randomised trials indicates that maintaining HbA1c within an optimal range significantly reduces the incidence of microvascular and macrovascular events, regardless of diabetes duration, with the most favourable outcomes observed at achieved HbA1c levels of 7.1-7.7% [3]. This relationship between glycaemia and treatment outcomes explains why interventions that bring HbA1c

within an appropriate target range - whether pharmacological or behavioural - are so central to diabetes management. It also provides the benchmark against which the value of resistance training is assessed.

Lifestyle modification, including diet, physical activity, and weight management, is the foundation of T2DM management. All major clinical guidelines recommend it as first-line management alongside pharmacotherapy [4]. Historically, exercise recommendations focused on aerobic activities such as walking, cycling, or swimming, and the evidence base for aerobic exercise in diabetes has long been established. Resistance training, defined as exercise in which muscles work against an external load to increase strength and muscle mass, has received relatively less research attention than aerobic training [5]. This is despite a strong physiological rationale for its use in a disease whose central defect is impaired muscular glucose uptake [5]. Current standards of care explicitly list resistance exercise, alongside aerobic activity, among first-line lifestyle interventions [4].

The physiological rationale for this recommendation is strong. Skeletal muscle is the largest insulin-sensitive tissue in the body and the principal site of postprandial glucose uptake; it accounts for most insulin-stimulated glucose disposal [5]. Age-related loss of muscle mass, or sarcopenia, therefore reduces the body's capacity to clear glucose from the bloodstream and aggravates insulin resistance, creating a vicious cycle that resistance training is well positioned to interrupt [6,7]. By increasing or preserving muscle mass and the density of the glucose transporter GLUT4, resistance training expands the tissue reservoir into which glucose can be taken up and improves the efficiency of this process. This mechanistic rationale is reinforced by a growing body of evidence from randomised trials [8,9].

Despite guideline endorsement and a clear mechanistic rationale, resistance training remains under-prescribed in routine diabetes care. This is partly because clinicians are less familiar with this form of activity than with aerobic exercise recommendations, and partly because of concerns about safety in older patients with comorbidities, even though resistance training is safe for most of this population when appropriate precautions are taken [2]. Given the multidisciplinary nature of diabetes management, involving physicians, diabetologists, physiotherapists, exercise physiologists, dietitians, and nurses, a shared and current understanding of the value, evidence, and practical aspects of resistance training is needed. This review addresses that need.

Research objective. To provide an integrated, up-to-date synthesis of the mechanisms, clinical efficacy, comparative and combined effects, training parameters, safety, and practical prescription of resistance training for glycaemic control in adults with T2DM, including considerations for implementation in the Polish setting.

Research questions. (1) Through which physiological mechanisms does resistance training improve glycaemic control in T2DM? (2) What is the magnitude and clinical significance of its effect on HbA1c and on related metabolic and functional outcomes? (3) How does resistance training compare with aerobic training, and what are the effects of combining the two modalities? (4) Which training parameters - frequency, intensity, and volume - are associated with glycaemic benefit, and how consistent is that evidence? (5) Is this form of activity safe in adults with T2DM, including those with diabetes-related complications? (6) How should it be prescribed in practice, and what would support its implementation in the Polish setting?

Research hypotheses. Questions 1, 4, 5 and 6 are descriptive and were addressed synthetically. For questions 2 and 3, which concern the direction and magnitude of effect, we hypothesised that: (i) resistance training produces clinically meaningful reductions in HbA1c in adults with T2DM; (ii) its effect on glycaemia is broadly comparable to that of aerobic training; (iii) combining the two modalities produces greater benefits than either alone; and (iv) resistance training confers additional metabolic and functional benefits, including gains in muscular strength, that are of particular value in older patients.

2. Research Materials and Methods

2.1. Study design and information sources

This narrative review did not involve human participants; it synthesised peer-reviewed literature on resistance training and glycaemic control in T2DM. PubMed/MEDLINE, Scopus, and the Cochrane Library were searched for publications from January 2007 through January 2026. The search strategy combined controlled terms and free-text terms: (“resistance training” OR “strength training” OR “weight training”) AND (“type 2 diabetes” OR “T2DM”) AND (“glycaemic control” OR “glycemic control” OR “HbA1c” OR “insulin sensitivity” OR “insulin resistance”).

2.2. Procedure

Evidence was selected purposively, with priority given to randomised controlled trials, systematic reviews, meta-analyses, and joint position statements and guidelines from the American Diabetes Association (ADA) and the American College of Sports Medicine (ACSM). The reference lists of key articles were hand-searched to identify additional sources. Only full-text English-language publications concerning adults with T2DM were included, except for a small number of sources cited solely to describe the implementation context. Titles and abstracts were reviewed by the authors for relevance to the stated eligibility criteria, and extracted findings were organised thematically according to mechanisms, efficacy, comparisons, training parameters, safety, and exercise prescription.

2.3. Data collection and analytical approach

Because the review was qualitative and integrative, no meta-analytic pooling or formal risk-of-bias assessment was undertaken. The findings were synthesised narratively and presented in summary tables covering mechanisms of benefit, principal randomised trials, comparisons between exercise modalities, and practical resistance-training recommendations.

3. Results - Literature Findings

3.1. Glycaemic dysregulation in type 2 diabetes: the central role of skeletal muscle

Insulin resistance in peripheral tissues is central to the pathophysiology of T2DM, and skeletal muscle is quantitatively the most important site of insulin-stimulated glucose disposal [5]. In healthy muscle, insulin promotes the translocation of GLUT4 transporters to the cell membrane, facilitating glucose uptake. In T2DM, impaired insulin signalling limits GLUT4 translocation and reduces muscular glucose uptake, contributing to hyperglycaemia [5]. Given the large mass of skeletal muscle, even partial restoration of its glucose-disposal capacity can substantially improve whole-body glycaemia. Increasing muscle glucose uptake - through contraction-mediated pathways or improved insulin signalling - is therefore a rational therapeutic target [5].

This muscle-centred perspective explains why loss of muscle mass and function is detrimental in diabetes. Sarcopenia is a skeletal muscle disorder characterised by low muscle strength and reduced muscle quantity or quality [6,10]; it reduces the capacity of a major glucose-disposal tissue and may aggravate insulin resistance [5]. Interventions that build or

preserve muscle therefore address a mechanistically relevant component of the disease - the central aim of resistance training. The bidirectional relationship between sarcopenia and T2DM, in which each condition can exacerbate the other, is well documented [6,10].

The clinical relevance of this cycle increases with population ageing. A growing proportion of adults with T2DM are older, and sarcopenia is common in this population [6,7]. Both muscle quantity and quality may be impaired: intramuscular fat infiltration, or myosteatosis, is increasingly recognised as a mechanistic link between ageing, T2DM, and sarcopenia, and together with mitochondrial dysfunction can further reduce glucose uptake [7]. Although aerobic exercise can improve muscle metabolism and function, it generally induces less hypertrophy than resistance training and therefore targets muscle loss less directly. Given the central role of muscle in glucose disposal and this bidirectional relationship, resistance training is particularly relevant to older adults with diabetes. Regular physical activity and nutritional support are increasingly recommended alongside standard glycaemic therapy to address associated muscle loss [10].

3.2. Mechanisms by which resistance training improves glycaemic control

Resistance training may improve glycaemia through several complementary mechanisms operating over different timescales. Acutely, muscle contraction increases insulin-independent glucose uptake, and a single resistance-exercise session may enhance post-exercise insulin sensitivity [5]. Chronically, repeated training can produce muscular hypertrophy that expands glucose-disposal capacity; increase GLUT4 content and translocation [5]; improve proximal insulin signalling and whole-body insulin sensitivity [5]; and improve body composition by preserving or increasing lean mass. Importantly, these adaptations are not contingent on hypertrophy: when glucose disposal is normalised to muscle mass, differences between resistance-trained and untrained individuals disappear, indicating that qualitative adaptations within muscle contribute independently of any gain in mass [5]. These mechanisms are summarised in Table 1.

Table 1. Principal mechanisms by which resistance training improves glycaemic control in T2DM

Mechanism	Effect on glucose metabolism
Increased or preserved muscle mass	Expands the principal reservoir for insulin-mediated glucose disposal [5].
Increased GLUT4 content/translocation	Enhances glucose transport into muscle for a given insulin signal [5].
Improved insulin signalling/sensitivity	Restores the efficiency of insulin-stimulated glucose uptake [5].
Acute contraction-mediated uptake	Insulin-independent glucose uptake during and after exercise; lowers post-exercise glycaemia [5,11].

Mechanism	Effect on glucose metabolism
Improved body composition	Preserves or increases lean body mass and may improve body composition [5].
Reduced sarcopenia	Preserves the principal glucose-disposal reservoir, which is particularly important in older patients [6,7,10].

Source: authors' synthesis based on references [5,6,7,10,11]. GLUT4 - glucose transporter type 4; T2DM - type 2 diabetes mellitus.

The acute and chronic effects operate through partly different pathways and both are clinically relevant. Because contraction-mediated uptake is insulin-independent, exercise can lower glycaemia even when insulin signalling is impaired; together with a period of heightened insulin sensitivity after a session, this accounts for the acute reduction in glucose that patients may observe. In a small randomised crossover study using continuous glucose monitoring, postprandially elevated glucose declined after a single 20-minute session of resistance exercise, although the study included only six participants and no non-exercising comparison condition [11]. Chronic adaptations raise the baseline capacity for glucose uptake, so regular training produces a sustained improvement in glycaemic control reflected in HbA1c [5,12]. In practice, the two types of effect are complementary: the acute response supports day-to-day glycaemic management, whereas the chronic response underlies the long-term reduction in HbA1c. Consistent with this, meta-regression has linked the number of repetitions per set to the magnitude of HbA1c change [12].

A further mechanistic dimension concerns the adipose tissue-muscle axis and chronic low-grade inflammation. Excess visceral and intramuscular fat is a source of pro-inflammatory adipokines and free fatty acids that disrupt insulin signalling, while the chronic subclinical inflammation characteristic of T2DM contributes to insulin resistance [13,14]. By improving body composition and skeletal muscle metabolism, resistance training may shift this balance favourably [5,13,14]. This anti-inflammatory effect may contribute to improvements in glycaemia and the overall cardiometabolic risk profile of people with diabetes. Two meta-analyses of randomised trials in people with T2DM each found a modest but significant reduction in C-reactive protein with supervised resistance training, whereas the pooled effects on tumour necrosis factor alpha and interleukin-6 did not reach statistical significance in either [13,14].

3.3. Efficacy of resistance training alone for glycaemic control

Evidence from randomised trials indicates that resistance training alone can improve glycaemic control in T2DM, although not every trial has demonstrated a significant effect. A meta-analysis of randomised trials of resistance training in patients with T2DM aged 60 years

and over found a significant reduction in HbA1c and a significant gain in muscular strength, whereas the pooled effect on lean body mass did not reach statistical significance; training intensity did not significantly moderate the effect on HbA1c in that analysis. These findings support the efficacy of the modality in older adults, a group in whom preservation of strength and function is particularly important [15]. A separate systematic review and meta-analysis of resistance training in T2DM found, in contrast, that higher-intensity protocols produced significantly greater reductions in HbA1c than low- or moderate-intensity training (approximately 0.6 versus 0.2 percentage points), together with significant reductions in insulin levels. A reduction of this magnitude is clinically relevant for patients whose HbA1c lies above the target range, given the established association between glycaemic control and complication risk [3]. Adequate training intensity is therefore recommended in physical activity guidance [16]. Selected studies, including those on comparative and combined training discussed in the following sections, are summarised in Table 2.

Table 2. Selected studies and evidence syntheses examining the effects of resistance and combined training on HbA1c in T2DM

Study (first author, year)	Design/comparison	Principal glycaemic finding
Lee, Kim and Kim, 2017 [15]	Meta-analysis of RCTs; resistance training versus control (older adults)	Significant reduction in HbA1c; pooled effect on lean body mass was not statistically significant.
Sigal et al., 2007 (DARE) [17]	Aerobic, resistance, and combined training versus control	Each training modality improved HbA1c; combined training was superior to either modality alone.
Church et al., 2010 (HART-D) [18]	Aerobic, resistance, and combined training versus control	Only combined training significantly reduced HbA1c compared with control.
Bacchi et al., 2012 (RAED2) [19]	Head-to-head comparison of aerobic and resistance training	Both modalities produced similar improvements in HbA1c and metabolic control.

Source: authors' synthesis of the cited studies. DARE - Diabetes Aerobic and Resistance Exercise; HART-D - Health Benefits of Aerobic and Resistance Training in Individuals with Type 2 Diabetes; HbA1c - glycated haemoglobin; RAED2 - Resistance Versus Aerobic Exercise in Type 2 Diabetes.

3.4. Resistance versus aerobic training

Head-to-head trials indicate that resistance and aerobic training produce broadly similar average improvements in glycaemic control. In the DARE trial, both resistance training alone and aerobic training alone improved HbA1c relative to the control group, with effects of similar magnitude (reductions of 0.38 and 0.51 percentage points, respectively) [17]. The RAED2 trial likewise found similar efficacy of resistance and aerobic training for metabolic control in adults with T2DM (reductions of 0.35 and 0.40 percentage points) [19]. This comparability is clinically useful, because resistance training may provide a viable alternative for people who cannot or do not wish to undertake prolonged aerobic activity owing to joint disease, obesity, or personal preference. Exercise modality can therefore be individualised while retaining the potential for clinically meaningful glycaemic benefit.

3.5. Combined resistance and aerobic training

Combined training often provides greater glycaemic benefit, although differences in total exercise dose and time allocation across trials should be considered. In the DARE trial, combined aerobic and resistance training produced greater improvements in HbA1c than either modality alone, and the authors recommended that patients seeking to improve metabolic control through exercise perform both types of training [17]. In the HART-D trial, in which total exercise time was matched across groups, only combined aerobic and resistance training significantly reduced HbA1c compared with the non-exercising control group [18]. These findings suggest that dividing a given exercise time between the two modalities may be advantageous.

At the level of pooled evidence, a large meta-analysis found that structured training - aerobic, resistance, or combined - was associated with an absolute HbA1c reduction of approximately 0.67 percentage points, with the greatest effect achieved by more than 150 minutes of structured exercise per week [20]. Within that analysis, however, the pooled subgroup estimates for aerobic, resistance and combined training were of broadly similar magnitude and did not show an advantage for combined training [20]. A more recent systematic review and meta-analysis of combined aerobic and resistance training in adults with T2DM and overweight or obesity reported improvements in glycaemic control, blood pressure and quality of life, including reductions in tumour necrosis factor alpha and interleukin-6 - markers that have not responded consistently to resistance training alone [13,21]. A subsequent meta-regression found that the weekly volume of resistance exercise explained heterogeneity within combined training, supporting the value of including resistance work in a combined programme [22].

Taken together, the evidence favours combined training where feasible: head-to-head trials show an advantage, whereas pooled subgroup estimates do not, and resistance training remains a core component of either approach.

3.6. Training parameters: frequency, intensity, and volume

Whether the glycaemic response to resistance training is dose-responsive remains unresolved, and the optimal dose is uncertain. A systematic review with meta-regression found that supervised exercise volume was associated with HbA1c change in aerobic training, but that no training variable correlated with the glycaemic response to resistance training performed

alone [22]. A more recent meta-analysis likewise found no association between the total number of training sessions completed and the change in HbA1c, and no moderating effect of intervention duration [23].

The evidence for intensity is more consistent: higher intensities (70-80% of one-repetition maximum [1RM]) have been associated with larger improvements in glycaemic control [2,12,16], and a dedicated dose-response meta-analysis found that repetitions per set independently predicted the magnitude of HbA1c reduction, with subgroup analyses indicating greater benefits after 12-16 weeks of training, at 70-80% 1RM, and with 2-3 sessions per week [12]. This association is not universal, however, and one meta-analysis in older patients found no significant effect of training intensity on HbA1c [15]. An earlier mechanistic review likewise concluded that a consistent regimen at moderate intensity elicits the most beneficial metabolic responses in terms of HbA1c and insulin sensitivity [5].

Taken together, the evidence is divergent: intensity appears more consistently related to glycaemic benefit than total training volume. Pending firmer dose-response data, guidelines and study protocols generally recommend resistance training on at least two, and preferably three, non-consecutive days per week. Training should involve the major muscle groups, be performed at moderate-to-vigorous intensity, and use progressive increases in load as strength improves [2], with progression kept gradual, particularly in older or less-trained adults. The available data therefore support a structured, progressive programme rather than unsystematic activity.

3.7. Additional metabolic and functional benefits

Beyond glycaemia, resistance training provides benefits of particular relevance to people with T2DM. It increases strength and can preserve or increase lean body mass, thereby addressing sarcopenia, a condition associated with poorer metabolic health [6,7]. Resistance training may improve body composition [5] and several cardiometabolic risk factors, including blood pressure [2,21], lipid profile [2], and markers of systemic inflammation [13,14,21]. It may also help preserve bone mineral density, an important consideration in a population at increased fracture risk [2,7]. Improvements in strength and mobility support broader healthy-ageing goals [2,7].

Table 3. Comparative effects of aerobic, resistance, and combined training in T2DM

Outcome	Aerobic training	Resistance training	Combined training
HbA1c reduction	Yes (approx. 0.5-0.7%)	Yes (approx. 0.3-0.6%)	Generally greatest; not superior in every pooled analysis
Insulin sensitivity	Improved	Improved	Improved
Muscle strength	Minimal change	Marked increase	Marked increase
Muscle mass	Minimal change	Preserved or increased; pooled effect not consistently significant	Preserved or increased
Cardiorespiratory fitness	Marked increase	Minimal change	Increased
Fat mass/body composition	Improved	Mixed evidence	Variable, depending on the study
Suitability for frail/older patients	Variable	Potentially high with individualisation and supervision	Potentially high if tolerated

Source: authors' synthesis based on references [9,16,17,18,19,20,23]. Values are indicative ranges, not precise point estimates. HbA1c - glycated haemoglobin.

Several of these additional benefits are directly relevant to outcomes that matter to people with diabetes. Because higher levels of physical activity are associated with lower cardiovascular incidence and mortality and with fewer microvascular complications in diabetes [4], the cardiometabolic improvements outlined above may translate into a more favourable overall risk profile. Preserving muscle strength and power has direct functional consequences in activities such as rising from a chair, climbing stairs, carrying loads, and maintaining balance. These gains may support independence and reduce fall risk in older adults, among whom diabetes frequently coexists with frailty [7,10]. Improvements in physical function, self-efficacy, mood, and quality of life extend the clinical relevance of resistance training beyond HbA1c alone [2,7]. For clinicians, these broader effects provide additional patient-centred reasons to recommend resistance training.

3.8. Safety, contraindications, and special populations

When appropriately prescribed and progressively advanced, resistance training is safe for the vast majority of patients with T2DM [2], although in the DARE trial adverse events, predominantly musculoskeletal, were more common in the exercise groups than in the non-exercising control group [17].

Several issues must nevertheless be considered. Patients with proliferative or severe non-proliferative retinopathy may need to avoid very high-intensity efforts and Valsalva manoeuvres because of the theoretical risk of retinal or vitreous haemorrhage; specialist advice should be sought in such cases [2]. Autonomic neuropathy, cardiovascular disease, and uncontrolled hypertension warrant particular caution and, where indicated, supervision [2,4].

Peripheral neuropathy and foot complications require attention to footwear and injury prevention [2]. Patients treated with insulin or insulin secretagogues should be counselled about the risk of exercise-related hypoglycaemia and the need to monitor glucose around training [2,4]. In older and untrained individuals, gradual introduction of the programme under competent supervision maximises benefit and minimises risk; importantly, this group may gain substantial benefits in function and independence. Most patients can therefore train safely and effectively despite having diabetes-related complications, provided these precautions are observed [2].

A risk-stratified pre-participation assessment supports safe implementation. For most individuals beginning low- to moderate-intensity activity such as brisk walking, no pre-exercise medical evaluation is needed unless symptoms of cardiovascular disease or microvascular complications are present, although clearance is recommended for currently sedentary adults before moderate- to high-intensity activity; notably, no evidence is available on whether pre-exercise stress testing is necessary or beneficial before resistance training specifically [2,9]. In practice, assessment should focus on cardiovascular risk, diabetes-related complications, symptoms, and functional status, with more detailed evaluation reserved for individuals with known cardiovascular disease, relevant symptoms, or plans to undertake vigorous training [2,4]. With these precautions, the benefit-risk balance is favourable, and excessive caution may unnecessarily restrict access to beneficial activity.

3.9. Practical recommendations

Translating the evidence into practice, pragmatic recommendations for resistance training in T2DM follow the familiar frequency, intensity, time, and type (FITT) framework and are consistent with ADA and ACSM guidance [2,8,9]. Training should be undertaken on two to three non-consecutive days per week [2,4]. Consensus guidance specifies one to three sets of 10-15 repetitions across 8-10 multi-joint exercises engaging the major muscle groups, at a moderate (50-69% 1RM) to vigorous (70-85% 1RM) intensity [2]. Dose-response data suggest that approximately three sets of 8-10 repetitions at 70-80% 1RM with short rest intervals are associated with the largest reductions in HbA1c [12]. Where possible, resistance training should be combined with aerobic activity in accordance with recommendations for both modalities [4]. The programme should be tailored to the patient's comorbidities, preferences, and baseline

fitness and, ideally, initially performed under the supervision of a qualified exercise professional.

The principle of progressive overload deserves particular emphasis because it is the element most often neglected when resistance training is prescribed in general terms. Muscle adapts to the demands placed upon it, so a load that was initially challenging ceases to provide an adequate stimulus once strength has increased. Without planned increases in resistance, progress eventually plateaus and a substantial part of the potential benefit is not realised [2]. A practical approach is to increase the load when the patient can comfortably complete the target number of repetitions with good technique, while keeping progression gradual enough to avoid injury or discouragement; in progressive training, resistance is increased first, followed by the number of sets and then training frequency [2]. The progression goal proposed in the earlier joint position statement - three sets of 8-10 repetitions across 8-10 exercises after six months - closely matches the parameters later identified in dose-response data [8,12]. At the same time, the programme should begin at a level appropriate to the individual's baseline capacity. An older patient with a low level of fitness may start with body-weight exercises, light resistance, and simple machines before advancing to more demanding forms, whereas a fitter individual may begin more ambitiously. Periodic reassessment allows the prescription to be adjusted as the patient progresses. This individualised, progressive approach is consistent with the training parameters associated with greater HbA1c reductions in T2DM [12]. These principles are summarised in Table 4.

Table 4. FITT-based resistance-training recommendations for adults with T2DM

FITT element	Recommendation
Frequency	2-3 non-consecutive days per week.
Intensity	Moderate (50-69% of one-repetition maximum) to vigorous (70-85%); dose-response data favour 70-80%.
Time/volume	1-3 sets of 10-15 repetitions per exercise, using 8-10 exercises engaging the major muscle groups; dose-response data favour approximately 3 sets of 8-10 repetitions with rest intervals under 60 seconds.
Type	Exercises engaging the major muscle groups, using free weights or machines; combine with aerobic training where feasible.
Progression/supervision	Increase resistance first, then sets, then frequency; keep progression gradual. Initial supervision, especially for older or higher-risk patients.

Source: authors' synthesis based on references [2,8,12]. FITT - Frequency, Intensity, Time, Type; T2DM - type 2 diabetes mellitus. Recommendations should be individualised and, where indicated, preceded by medical evaluation.

3.10. Resistance training within the wider treatment context

Resistance training does not operate in isolation but is one component of a comprehensive management strategy that also includes nutrition, weight management, aerobic activity,

pharmacotherapy, and the reduction of sedentary behaviour. Its relationship with these other elements is complementary rather than competitive [4]. This complementary relationship means that exercise can support the effects of pharmacotherapy and that, in some patients, glucose-lowering treatment may be modified as glycaemic control improves; any such change must be introduced by the treating physician with appropriate monitoring, particularly in patients using insulin or insulin secretagogues [2,4]. Findings from combined-training studies support this complementary approach: concurrent aerobic and resistance training improves glycaemic, vascular, and quality-of-life outcomes together [21]. Reducing and interrupting prolonged sitting, for example with brief periods of activity, has also emerged as an independent target that complements structured training, with current standards advising that prolonged sitting be interrupted at least every 30 minutes [4,9].

3.11. Adherence and behaviour change

The glycaemic benefits of resistance training depend on sustained participation, and adherence remains a major challenge in clinical practice. Initial supervision can help develop skills and confidence, and counselling by health care professionals is a recognised source of support; realistic, individualised goals, gradual progression that minimises early injury or discouragement, and integration of training into established routines may also help [2]. Beyond the evidence reviewed above, self-monitoring that demonstrates individual glycaemic responses, framing the intervention in terms of everyday function rather than glycaemic values alone, and family or group involvement are plausible but less well-studied strategies.

Evidence for home-based training is mixed. A systematic review and meta-analysis found a significant HbA1c reduction of approximately 0.4 percentage points in supervised gym-based resistance-training programmes, whereas the pooled effect of minimally equipped home-based programmes was not statistically significant [23]. The same pattern was noted in earlier consensus guidance, which observed that home-based training after supervised gym-based training may be less effective for glycaemic control while remaining adequate for maintaining muscle mass and strength [8]. This supports structured supervision but does not establish that home-based training is ineffective, and home-based options may still improve access and overall physical activity.

4. Discussion

The evidence synthesised in this review broadly supports the four review hypotheses, with important qualifications. Resistance training is associated with clinically meaningful average reductions in HbA1c in adults with T2DM [15,16,24]; its average effect on glycaemia is broadly comparable to that of aerobic training [17,19]; combining the two modalities generally provides greater benefit than either alone [17,18]; and resistance training offers additional metabolic and functional benefits of particular value in older adults, including gains in muscular strength [2,7,15]. These are average effects rather than uniform findings: effects on lean mass are less consistent than those on strength, and in the HART-D trial neither resistance nor aerobic training alone significantly reduced HbA1c relative to control, while only combined training did [18].

The overall pattern of findings across mechanistic studies, randomised trials, and pooled analyses nevertheless supports guideline recommendations to include resistance training in exercise therapy for diabetes [4,8,9]. Across pooled analyses, average reductions in HbA1c with resistance training cluster between roughly 0.3 and 0.6 percentage points, with smaller effects at low training intensities and larger effects at higher intensities [16,20,23,24].

Several implications for clinical practice follow. First, broadly similar average effects of resistance and aerobic training allow modality selection to be guided by individual preferences, abilities, and comorbidities [17,19]. Obesity and knee or hip osteoarthritis are recognised barriers to physical activity participation [2]; for such patients, and for those with a strong aversion to prolonged aerobic activity, resistance training may offer a more acceptable route to glycaemic benefit. Second, the HART-D trial suggests that dividing a fixed exercise time between aerobic and resistance training may provide greater glycaemic benefit than allocating it to one modality alone [18]. Third, because skeletal muscle is central to glucose disposal and sarcopenia is common in older adults with T2DM, resistance training has particular metabolic and functional relevance in this population [5,6,7].

4.1. Practical Implications

The following composite, hypothetical scenario, not based on the history of any individual patient, illustrates how the principles discussed above translate into practice. A man in his early sixties with T2DM of several years' duration, treated with metformin, presents for a routine follow-up appointment. His HbA1c is 7.8%; he has central obesity, reduced grip strength, and

a sedentary lifestyle. Knee osteoarthritis makes prolonged walking uncomfortable, so previous advice to “do more cardio” has been unsuccessful. The diabetes team recognises that resistance training may provide a glycaemic improvement comparable to aerobic exercise while being better tolerated by his knees and countering early sarcopenia. After a review of cardiovascular risk and symptoms, he is therefore referred to a physiotherapist for a supervised, progressive resistance-training programme on two to three days per week, with low-impact aerobic activity added later as his tolerance improves. Over the following months, he gains strength, reports greater ease in performing daily activities, and his HbA1c improves to a clinically meaningful extent. Glucose monitoring shows him the effect of training and reinforces adherence. The scenario reflects the central themes of the review: the mechanistic importance of targeting muscle, the value of individualising exercise modality, the potential added benefit of combining resistance and aerobic training, and the role of supervision and self-monitoring in adherence.

A recurring practical barrier is that resistance training remains under-prescribed relative to the evidence supporting it. Contributing factors may include limited clinician familiarity, the absence of exercise professionals in many diabetes services, patients’ apprehension about “weight training”, and perceived safety concerns. Addressing these barriers requires clinician endorsement, structured referral pathways to physiotherapists and exercise physiologists, and education that presents resistance training as accessible and adaptable.

These considerations are directly relevant to the Polish clinical setting. The 2025 Diabetes Poland recommendations advise people with T2DM to add resistance exercise to aerobic activity [25]. Practical priorities include brief exercise counselling during routine diabetes visits, links between diabetology, sports medicine, and physiotherapy, and access to supervised programmes for older and higher-risk adults. Existing physiotherapy, community, and outpatient rehabilitation services may support delivery, but implementation will depend on reimbursement, the availability of appropriately trained exercise professionals, and integration of structured physical activity into diabetes education. In a nationally representative Polish survey, competing priorities, low motivation, fatigue, and low energy were prominent perceived barriers to regular physical activity [26]. A survey of patients with type 2 diabetes in Poland similarly found that few intensified their physical activity after diagnosis and that many did not regard exercise as an important component of treatment [27]. Implementation therefore remains a major challenge.

4.2. Limitations

The limitations of this review should be acknowledged. As a narrative rather than systematic review, it is susceptible to selection bias. Restriction to full-text English-language publications may also have excluded relevant evidence, and no formal risk-of-bias assessment or quantitative synthesis was undertaken. Prioritising randomised trials, meta-analyses, and consensus guidelines partly mitigated these limitations. Heterogeneity in training intensity, volume, duration, and supervision, as well as in patient characteristics such as baseline HbA1c, complicates direct comparisons. Reported HbA1c effect sizes should therefore be interpreted as indicative ranges rather than precise estimates. These caveats reduce the precision and certainty of the conclusions, although the overall direction of effect is consistent across multiple study types.

4.3. Future Directions

Several questions warrant further investigation. The optimal dose of resistance training - the balance of intensity, volume, and frequency required to maximise glycaemic benefit while maintaining adherence - has not been definitively established. The relative merits of higher-intensity, lower-volume protocols compared with more moderate approaches require further study, particularly in older patients. The durability of benefit after supervised programmes end and the strategies that most effectively sustain long-term engagement are of central practical importance. Most studies have used surrogate endpoints such as HbA1c rather than hard clinical endpoints. Longer studies linking resistance training to complications, cardiovascular events, and mortality would therefore strengthen the evidence base.

A further priority is implementation research examining how best to incorporate resistance-training prescription and referral into routine diabetes services in specific healthcare systems, including Poland, in order to reduce the gap between what the evidence supports and what patients actually receive. Scalable, lower-cost delivery models, including home-based programmes, deserve particular attention. Current evidence is more consistent for HbA1c reduction with supervised gym-based resistance training than with minimally equipped home-based programmes, but direct comparative evidence remains limited [23]. Future research should determine whether remote supervision, higher intensity, or more frequent professional contact can improve the effectiveness and reach of home-based training.

5. Conclusions

Resistance training is an effective, generally safe, and guideline-endorsed strategy for improving glycaemic control in adults with T2DM. By increasing or preserving skeletal muscle mass, enhancing GLUT4-mediated glucose uptake and insulin sensitivity, and improving body composition, it targets a central pathophysiological component of T2DM. Average reductions in HbA1c are broadly comparable to those achieved with aerobic training, while combined resistance and aerobic training generally provides the greatest glycaemic benefit. These are average effects, and individual trials have not uniformly shown a significant benefit from either modality alone.

For the multidisciplinary diabetes team, the practical priorities are clear: recognise resistance training as a core therapy rather than an optional extra; prescribe it on two to three days per week at an adequate, progressively increasing intensity, individualised to the patient and combined with aerobic activity where possible; attend to the specific safety needs of older and higher-risk patients while recognising that these groups often derive substantial benefit; and address the implementation barriers - insufficient clinician familiarity, gaps in referral pathways, and adherence problems - that currently limit its use. Physician endorsement and structured access to exercise professionals are the most immediate means of translating this evidence into better treatment outcomes. Further research should refine the optimal dose and long-term durability of benefit, extend the evidence to hard clinical endpoints, and evaluate strategies that increase the uptake and maintenance of activity in routine care, including in Poland. Embedding this evidence into everyday practice remains the central challenge.

Disclosures

Supplementary Materials

Not applicable.

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